Qualcomm Technologies, Inc.

# WLAN Conformance Testing and Compliant Power Configuration for 802.11ax Chipsets

## Information to Share with Test Lab

80-YB952-3 Rev. D

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# Revision History

| Revision | Date | Description |
|---|---|---|
| A | January 2018 | Initial release |
| B | February 2019 | Added *FCC Conformance Testing Guidance for DL OFDMA* |
| C | April 2019 | Added NOTE about Test tone XML files. |
| D | December 2019 | ▪ Updated 11ax Access Point guidance.<br>▪ Added 11ax Client device guidance. |

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# 1 Introduction

## 1.1 Purpose

This document provides customers and test labs with the necessary guidance and instructions to perform radio conformance for Qualcomm Technologies, Inc. (QTI) 802.11ax WLAN chipsets.

NOTE: This document is confidential and proprietary. However, manufacturers with an existing Technology License Agreement (TLA) or equivalent agreement ("Agreement") may share the information in this document with their chosen test lab subject to the terms of such Agreement. Manufacturers must maintain company confidential handling and marking of this technical documentation when sharing with their test lab.

Where required, only Chapter 7.1~~7~~ and Section 8.1 may be included in the formal test reports. Instructions to the test lab are included at the beginning of those sections.

Questions regarding the use of this confidential technical document should be directed to QTI Customer Engineering.

## 1.2 Scope

This document is applicable to QCA99XX, QCA98XX Access Point, and QCA629X Client 802.11ax chipsets. However, much of this document is informative for 802.11ac solutions.

- Chapter 2 is tutorial information on the QTI design to ensure compliant operation.
- Chapter 3 provides a summary of the supported 802.11ax single user and multi-user packet types for Qualcomm AP and Client chipsets.
- Chapter 4 contains guidance on FCC conformance testing for DL OFDMA.
- Chapter 5 contains guidance on FCC conformance testing for UL OFDMA.
- Chapter 6 contains guidance on operating modes for UL OFDMA.
- Chapter 7 contains a justification of worst-case operating modes for conformance testing, suitable for the lab to incorporate in the formal test reports.
- Chapter 8 describes the conformance test plan prepared by the manufacturer including a checklist of supported operating modes for the device under test.
- Chapter 9 contains a suggested test flow for the lab to follow to optimize testing. This test flow is important for 4x4 or 8x8 access points which pose a significant test burden.
- Chapter 10 contains guidance on using the Qualcomm Radio Control Toolkit (QCRT) test software to support RF conformance testing.


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# 2 Solution for Regulatory Compliant Powers – Tutorial

## 2.1 Introduction

This chapter provides an overview of the QTI design for globally-compliant operation.

NOTE: This document is for use by the conformance test lab.
The information in this chapter must not be provided to the FCC and must not be included in the formal test reports.

## 2.2 Conformance test limit overview

1. The CTL Table is stored in the product board data file (BDF) during manufacturing and is always *product-specific*. CTL data must be determined from formal conformance testing of each end-product and reflects each unique board/radio/antenna design.
2. A product's CTL file contains data for 3 or more regulatory regions (typically FCC, ETSI, and Japan/MKK). Later designs support the option of populating additional CTL groups for China and Korea.
3. The CTL file contains compliant power levels specific to 2.4 and 5 GHz channels, each modulation type, channel bandwidth and MIMO mode.
4. The CTL data takes into account the highest gain antennas to be used with the end product.
5. The CTL data contains per-chain conducted power limits for each operating mode.
6. The manufacturer should provide a preliminary CTL file at the start of the formal conformance testing project. This preliminary table indicates all supported operating modes and channels as well as preliminary target powers.


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## 2.3 Conformance test limit groups

**NOTE:** The QTI solution arranges all 802.11 modulations into the following CTL groups due to similarity in achievable compliant transmit powers. By design the same transmit powers are shared by all modulations/rates within each CTL group below.

**Table 2-1 Packet types included in each CTL group, ACCESS POINT Chipsets**

| | CTL Groups | | 5GHz Modes Included in Each Group | |
|---|---|---|---|---|
| | Name Used in Software | | 802.11bgn/ac | 802.11ax |
| 5.1 | 5G 11a (also referred to as "legacy") | | 11a | |
| 5.2 | 5G HT20/VHT20 | | HT20, VHT20 | HE20_SU and DL MU-MIMO |
| 5.3 | 5G DL_OFDMA_20 (Added for HK2.0) | | | DL-OFDMA 20MHz capability bandwidth |
| 5.4 | 5G HT40/VHT40 | | HT40, VHT40 | HE40_SU and DL MU-MIMO |
| 5.5 | 5G DL_OFDMA_40 (Added for HK2.0) | | | DL-OFDMA 40MHz capability bandwidth |
| 5.6 | 5G VHT80 | | HT80, VHT80 | HE80_SU and DL MU-MIMO |
| 5.7 | 5G DL_OFDMA_80 (Added for HK2.0) | | | DL-OFDMA_80 (and 160) capability bandwidth |
| 5.8 | 5G 80p80 (primary lower than secondary) | | VHT80+80 | |
| 5.9 | 5G 80p80 (primary higher than secondary) | | VHT80+80 | |
| 5.10 | 5G 160 | | VHT160 | HE160_SU and DL MU-MIMO |
| | | | | |
| | | | 2GHz Modes Included in Each Group | |
| | | | 802.11bgn | 802.11ax |
| 2.1 | 2G 11b (also referred to as "CCK") | | 11b | |
| 2.2 | 2G Legacy (11g) | | 11g | |
| 2.3 | 2G HT20/VHT20 | | HT20, VHT20 | HE20_SU and DL MU-MIMO |
| 2.4 | 2G DL_OFDMA_20 (Added for HK2.0) | | | DL-OFDMA 20MHz capability bandwidth |
| 2.5 | 2G HT40/VHT40 | | HT40, VHT40 | HE40_SU and DL MU-MIMO |
| 2.6 | 2G DL_OFDMA_40 (Added for HK2.0) | | | DL-OFDMA 40MHz capability bandwidth |

**NOTE:** CTL data is shared for legacy HT/VHT and 11ax HE_SU and 11ax DL_MU-MIMO waveforms for each supported bandwidth.

**NOTE:** Separate CTL groups for DL-OFDMA_20, 40, 80  80 are added for HK2.X and later chipsets (DL-OFDMA 80 and 160 share the same CTL group due to memory limitations)

**NOTE:** UL-OFDMA RU26, 52, 106, 242, 484 CTL groups are not used in Access Point designs


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## Table 2-2 Packet types included in each CTL group, CLIENT Chipsets

| | CTL Groups | 5GHz Modes Included in Each Group | |
|---|---|---|---|
| | **Name Used in Software** | **802.11bgn/ac** | **802.11ax** |
| 5.1 | 5G 11a (also referred to as "legacy") | 11a | |
| 5.2 | 5G HT20/VHT20 | HT20, VHT20 | HE20_SU and UL MU-MIMO |
| 5.3 | 5G HT40/VHT40 | HT40, VHT40 | HE40_SU and UL MU-MIMO |
| 5.4 | 5G VHT80 | HT80, VHT80 | HE80_SU and UL MU-MIMO |
| 5.5 | 5G 160 | VHT160 | HE160_SU and UL MU-MIMO (future) |
| 5.6 | 5G 11ax UL-OFDMA RU26 | | UL-OFDMA RU26 (20 and 40MHz preambles) |
| 5.7 | 5G 11ax UL-OFDMA RU52 | | UL-OFDMA RU52 (20 and 40MHz preambles) |
| 5.8 | 5G 11ax UL-OFDMA RU106 | | UL-OFDMA RU106 (20 and 40MHz preambles) |
| 5.9 | 5G 11ax UL-OFDMA RU242 | | UL-OFDMA RU242 (20 and 40MHz preambles) |
| 5.1 | 5G 11ax UL-OFDMA RU484 | | UL-OFDMA RU484 (40MHz preamble) |
| 5.11 | 5G 11ax UL-OFDMA RU996 | | UL-OFDMA RU996 (80MHz preamble) |
| 5.12 | 5G 11ax UL-OFDMA RU996*2 | | UL-OFDMA RU996 (160MHz preamble) - May be supported in future |
| | | | |
| | | **2GHz Modes Included in Each Group** | |
| | | **802.11bgn** | **802.11ax** |
| 2.1 | 2G 11b (also referred to as "CCK") | 11b | |
| 2.2 | 2G Legacy (11g) | 11g | |
| 2.3 | 2G HT20/VHT20 | HT20, VHT20 | HE20_SU and UL MU-MIMO |
| 2.4 | 2G HT40/VHT40 | HT40, VHT40 | HE40_SU and UL MU-MIMO |
| 2.5 | 2G 11ax UL-OFDMA RU26 | | UL-OFDMA RU26 (20MHz preamble) |
| 2.6 | 2G 11ax UL-OFDMA RU52 | | UL-OFDMA RU52 (20MHz preamble) |
| 2.7 | 2G 11ax UL-OFDMA RU106 | | UL-OFDMA RU106 (20MHz preamble) |
| 2.8 | 2G 11ax UL-OFDMA RU242 | | UL-OFDMA RU242 (20MHz preamble) |
| 2.9 | 2G 11ax UL-OFDMA RU484 | | UL-OFDMA RU484 (40MHz preamble) |

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**NOTE:** CTL data is shared for legacy HT/VHT and 11ax HE_SU and 11ax UL_MU-MIMO waveforms for each supported bandwidth.

**NOTE:** DL-OFDMA CTL groups are not used in Client designs


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## 2.3.1 CTL structure example

Table 2-3 illustrates the data in each CTL Group.

See *CTL Assistant Tool IPQ807X Application Note* (80-YB215-8) for additional tutorial for the CTL Scheme.

Also see the product-specific *Board Data File Application Note* for detailed CTL file structure for AP and Client products.

**Table 2-3 Data contained in each CTL Group**

| Group | Reg Domain | Beamforming | Mode | Num Chains on bit Mask | Num Streams on bit Mask | Power levels are per chain (dBm) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **5GHz VHT80/HE80** | **FCC channel list** | | | | | **5210** | **5290** | **5530** | **5610** | **5690** | **5775** |
| | FCC(0x10) | 0 | HT80 | 00000010 | 00000011 | 20.5 | 20.5 | 20.5 | 20.5 | 20.5 | 26.5 |
| | FCC(0x10) | 0 | HT80 | 00000001 | 00000001 | 24.0 | 24.0 | 24.0 | 24.0 | 24.0 | 30.0 |
| | FCC(0x10) | 1 | HT80 | 00000010 | 00000001 | 20.0 | 20.0 | 19.5 | 19.5 | 19.5 | 26.0 |
| | **ETSI channel list** | | | | | **5210** | **5290** | **5530** | **5610** | | |
| | ETSI(0x30) | 0 | HT80 | 00000010 | 00000011 | 16.0 | 16.0 | 15.5 | 15.5 | | |
| | ETSI(0x30) | 0 | HT80 | 00000001 | 00000001 | 19.5 | 19.5 | 19.0 | 19.0 | | |
| | ETSI(0x30) | 1 | HT80 | 00000010 | 00000001 | 13.0 | 13.0 | 12.5 | 12.5 | | |
| | **MKK channel list** | | | | | **5210** | **5290** | **5530** | **5610** | | |
| | MKK(0x40) | 0 | HT80 | 00000010 | 00000011 | 16.0 | 16.0 | 22.5 | 22.5 | | |
| | MKK(0x40) | 0 | HT80 | 00000001 | 00000001 | 19.0 | 19.0 | 25.5 | 25.5 | | |
| | MKK(0x40) | 1 | HT80 | 00000010 | 00000001 | 16.0 | 16.0 | 22.5 | 22.5 | | |

NOTE: New designs may support optional additional CTL Groups for China (CHN) and Korea (KOR)

Additional groups can be implemented (for 2.4 and 5 GHz) to assign compliant powers to any combination of 11ac spatial streams, fewer enabled transmit chains and beamforming enabled for chosen modes. The manufacturer's test plan will detail the planned configurations so the test lab performs all necessary compliance testing and returns the desired sets of CTL values.


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## 2.3.2 Stream mask and chain mask in CTL file

The following concepts should be reviewed to correctly interpret and create CTL files.

The bit mask scheme allows creation of any desired combination of separate CTL data for one or more groups of Nss or enabled chains. The scheme is completely configurable by the manufacturer, but the CTL Assistant tool will only generate allowable configurations.

### Spatial Stream Mask

The spatial stream mask specifies the spatial stream(s) to which the CTL entry applies. Legacy solutions up to 802.11ac support up to 4 bits. Next generation 802.11ax supports up to 8 bits.

More than one bit set to 1 means the same CTL entry (i.e., same power limit) is applied to each of the spatial streams identified by the mask:

**802.11ac Stream Mask examples:**

0001 - CTL Entry applies to Nss=1
0010 - CTL Entry applies to Nss=2
1000 - CTL Entry applies to Nss=4
1111 - CTL Entry applies to Nss=1, 2, 3, and 4 (i.e., the same power limit applied to all)

**802.11ax Stream Mask examples:**

0000 0001 - CTL Entry applies to Nss=1
1000 0000 - CTL Entry applies to Nss=8
1000 1011 - CTL Entry applies to Nss=8, 4, 2, and 1
1111 1111 - CTL Entry applies to Nss=1 to 8

(Any combinations of bits may be set, these are just a few examples.)

### Chain Mask

Chain mask specifies the configuration of enabled transmit chains/antennas to which the CTL entry applies. Legacy solutions up to 802.11ac support up to 4 chains/antennas. Next generation 802.11ax supports up to 8.

More than one bit set to 1 means the same CTL entry (i.e., the same power limit) is applied to configuration of enabled chains identified by the mask:

**802.11ac Chain Mask examples:**

0001 - CTL Entry applies to 1 chain operation
0010 - CTL Entry applies to 2 chain operation
1000 - CTL Entry applies to 4 chain operation
1111 - CTL Entry applies to 4, 3, 2, and 1 chain operation

(Any combinations of bits may be set. These are just a few examples.)

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# 3 Supported 802.11ax Packet Types

Supported 802.11ax single user and multi-user packet types for Qualcomm AP and Client chipsets are summarized below. The following data is current as of the publication date of this document.

### Table 3-1 HE_SU supported packets, Access Point and Client chips

| (All use Legacy Preamble of same BW as RU) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | LEGACY PREAMBLE (MHz) | RU26 | RU52 | RU106 | RU242 | RU484 | RU996 | RU996*2 |
| 2.4GHz | 20 | Not supported | Not supported | Not supported | Supported (HE_SU_20) | Not supported | Not supported | Not supported |
| | 40 | Not supported | Not supported | Not supported | Not supported | Supported (HE_SU_40) | Not supported | Not supported |
| 5GHz | 20 | Not supported | Not supported | Not supported | Supported (HE_SU_20) | Not supported | Not supported | Not supported |
| | 40 | Not supported | Not supported | Not supported | Not supported | Supported (HE_SU_40) | Not supported | Not supported |
| | 80 | Not supported | Not supported | Not supported | Not supported | Not supported | Supported (HE_SU_80) | Not supported |
| | 160 | Not supported | Not supported | Not supported | Not supported | Not supported | Not supported | Supported (HE_SU_160) |

### Table 3-2 UL OFDMA preamble and RU supported packets, Client chips only

| All use Legacy Preamble with larger BW than the assigned RU) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | LEGACY PREAMBLE (MHz) | RU26 | RU52 | RU106 | RU242 | RU484 | RU996 | RU996*2 |
| 2.4GHz | 20 | Supported | Supported | Supported | Supported | Not supported | Not supported | Not supported |
| | 40 | Not supported | Not supported | Not supported | Not supported | Supported | Not supported | Not supported |
| 5GHz | 20 | Supported | Supported | Supported | Supported | Not supported | Not supported | Not supported |
| | 40 | Supported | Supported | Supported | Supported | Supported | Not supported | Not supported |
| | 80 | Not supported | Not supported | Not supported | Not supported | Not supported | Supported | Not supported |
| | 160 | Not supported | Not supported | Not supported | Not supported | Not supported | Not supported | Supported |


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## Table 3-3 DL OFDMA preamble and RU supported packets

| | (All use Legacy Preamble with same BW as packet BW) | | | | |
|---|---|---|---|---|---|
| | LEGACY PREAMBLE (MHz) | DL_OFDMA_20 | DL_OFDMA_40 | DL_OFDMA_80 | DL_OFDMA_160 |
| 2.4GHz | 20 | Supported | Not supported | Not supported | Not supported |
| | 40 | Not supported | Supported | Not supported | Not supported |
| 5GHz | 20 | Supported | Not supported | Not supported | Not supported |
| | 40 | Not supported | Supported | Not supported | Not supported |
| | 80 | Not supported | Not supported | Supported | Not supported |
| | 160 | Not supported | Not supported | Not supported | Supported |
| Any combinations of these RU sizes Assigned -> | | RU26, RU52, RU106 | RU26, RU52, RU106, RU242 | RU26, RU52, RU106, RU242, RU484 | RU26, RU52, RU106, RU242, RU484, RU996 |

-Current QTI AP chipsets will assign and modulate 100% of the DL OFDMA packet bandwidth with some combination of RU's.
- Future QTI chipsets will allow partial RU allocation. Those future chipsets may leave approximately 50% of the DL OFDMA packet bandwidth unmodulated (ie RU's unassigned). - QC AP's do not support 1 full BW RU in DL OFDMA packets.

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# 4 Conformance Test Guidance for DL OFDMA (Access Point Use Case)

NOTE: This document is for use by the conformance test lab.
The information in this chapter must not be provided to the FCC and must not be included in the formal test reports.

NOTE: Access Point products that also support Client device use cases (i.e., AP that can transmit UL OFDMA packets) must also conform to test guidance in Chapter 5.

**Guidance for 11ax Access Point Chipsets**

The test lab must perform PSD, Power, Band Edge and Spurious testing for the HE_SU and DL OFDMA (fully populated) along with all legacy packet types and supported bandwidths. Refer to the tables in Chapter 3 for currently supported HE_SU and DL OFDMA packet types.

The 100% modulated DL OFDMA waveforms listed in table 4.1 should be used for formal conformance testing of current generation of QTI AP chipsets:

- OFDMA_HE20_Full
- OFDMA_HE40_Full
- OFDMA_HE80_Full

NOTE: Section 4.1 only applies to *future* AP chipsets when partially populated DL OFDMA packets are implemented.

## 4.1 Guidance for future AP chipsets that support DL OFDMA partial loading

NOTE: The FCC guidance and required testing for partially loaded DL OFDMA packets does not apply to current generation of QTI Access Point chipsets. Current chipsets will fully load each DL OFDMA packet and partially loaded packets will not occur during normal operation (i.e., mission mode).

**Therefore the following test instructions and partially modulated DL OFDMA waveforms will apply to *future generations of QTI AP chipsets when partial loading is implemented*.**

---

FCC Test Guidance for Devices that Support Partially Modulated DL OFDMA Packets:

On October 3, 2018, the FCC published *Interim Guidance for conformance testing of DL OFDMA Power Spectral Density*. The guidance is summarized as follows:


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- Perform all in band Conducted Power and PSD tests using one of the lower MCS values (for example, MCS 0) and any one GI. No need to repeat these tests with multiple MCS or GI values.
- Measure PSD under fully loaded configurations ($PSD_{full}$) and partially loaded configurations ($PSD_{par}$). For partially loaded configurations test using a center (contiguous) tone plan and an edge (non-contiguous) tone plan.
- Determine $\delta = PSD_{par} - PSD_{full}$.
- If $\delta > 0$ and $PSD_{par}$ can be reduced to $PSD_{full}$ all testing can be done under fully loaded conditions. Spot check for band edges under partial configurations.
- Otherwise, test both fully loaded and partially loaded configurations in their entirety.

---

To facilitate testing of partially-populated DL OFDMA tone plans per FCC guidance, QTI supplies a set of DL OFDMA tone plans for 20 MHz, 40 MHz, and 80 MHz bandwidths. These OFDMA test tone plans are:

- Fully modulated tone plans
  - OFDMA_HE20_Full
  - OFDMA_HE40_Full
  - OFDMA_HE80_Full
- Partially modulated tone plans with middle of the band unmodulated, i.e., edge (non-contiguous) tone plan
  - OFDMA_HE20_M_unmod
  - OFDMA_HE40_M_unmod
  - OFDMA_HE80_M_unmod
- Partially modulated tone plans with middle of the band modulated, while upper and lower band edge unmodulated, i.e., center (contiguous) tone plan
  - OFDMA_HE20_BE_unmod
  - OFDMA_HE40_BE_unmod
  - OFDMA_HE80_BE_unmod

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**Table 4-1 OFDMA test tone description**

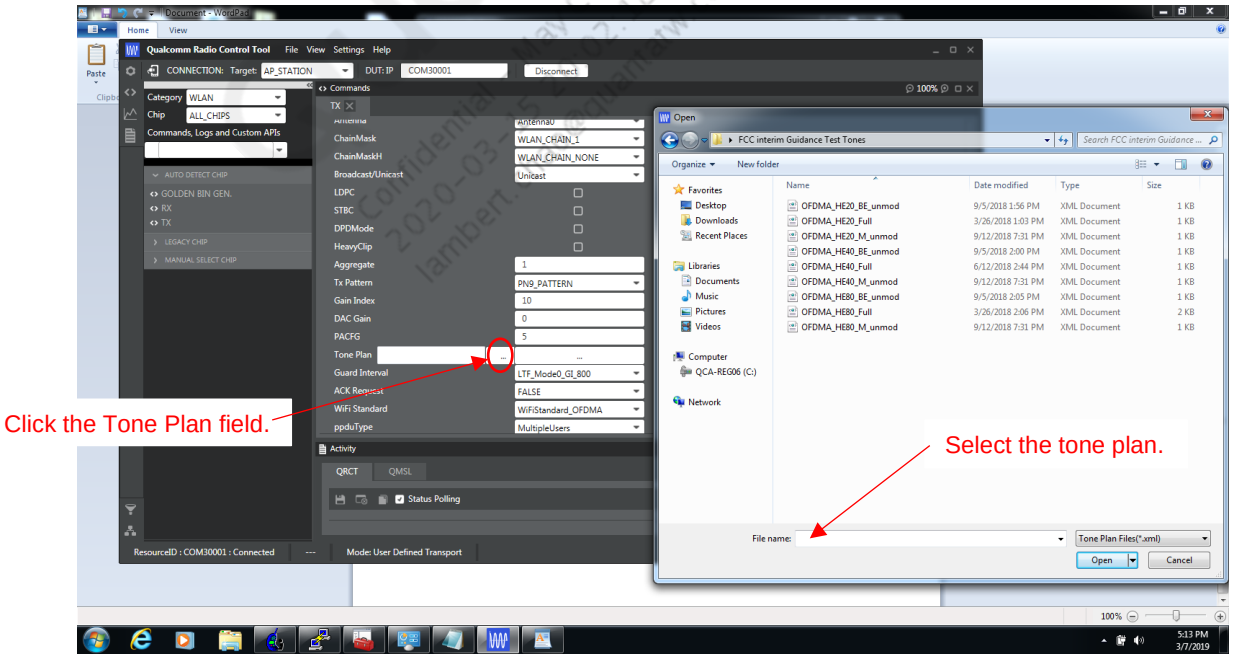
| OFDMA tone | Bandwidth | Tone plan* | % Modulation |
|---|---|---|---|
| OFDMA_HE20_Full | 20 MHz | 26_26_26_26_26_52_52 | 100% |
| OFDMA_HE20_M_unmod | 20 MHz | 26_26_26_26_26_26_26_26_26 | 66% |
| OFDMA_HE20_BE_unmod | 20 MHz | 26_26_26_26_26_52_52 | 56% |
| OFDMA_HE40_Full | 40 MHz | 52_52_26_106_52_52_26_106 | 100% |
| OFDMA_HE40_M_unmod | 40 MHz | 106_26_106_106_26_106 | 60% |
| OFDMA_HE40_BE_unmod | 40 MHz | 52_52_26_52_52_52_52_26_52_52 | 56% |
| OFDMA_HE80_Full | 80 MHz | 996 | 100% |
| OFDMA_HE80_M_unmod | 80 MHz | 242_242_26_242_242 | 48% |
| OFDMA_HE80_BE_unmod | 80 MHz | 106_26_106_106_26_106__26_106_26_106_106_26_106 | 72% |

*Red = Unmodulated RU

NOTE: Test tone XML files to support FCC testing using QSPR and QRCT are available for the WLAN AP software release in the FW package. The file is available at "/bdfUtil".

The following instructions describe how to import the DL OFDMA tone plans in QRCT-connectivity to perform FCC tests per the FCC guidance.

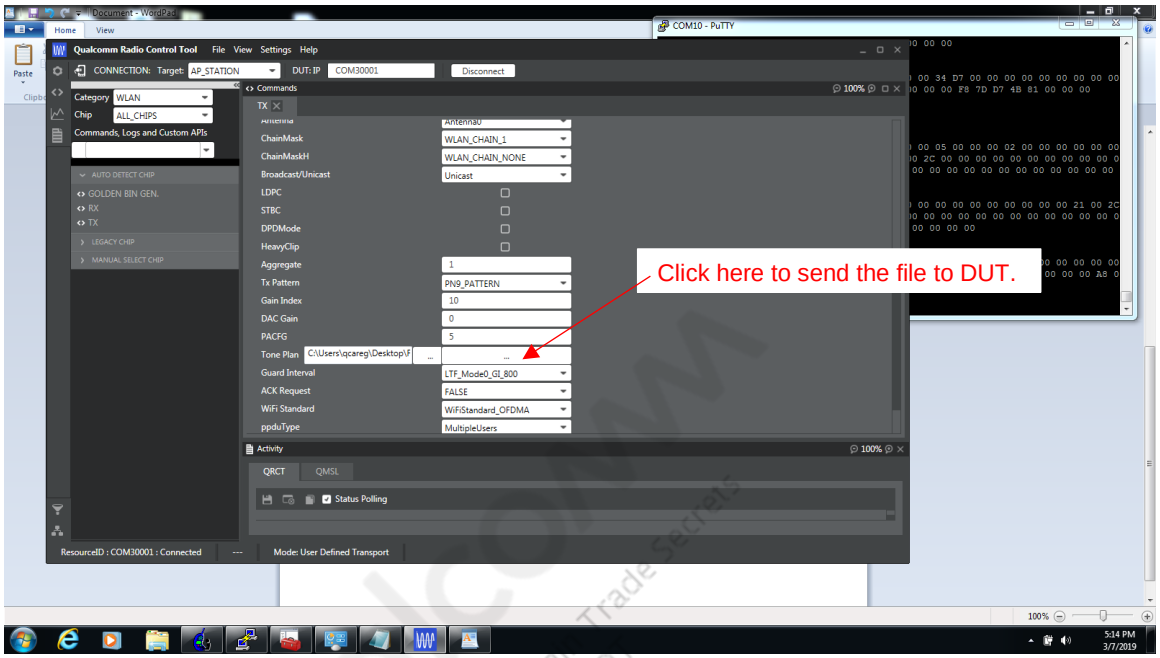
1. Click the Tone Plan field to display the list of tone plans. Select the plan to be tested.



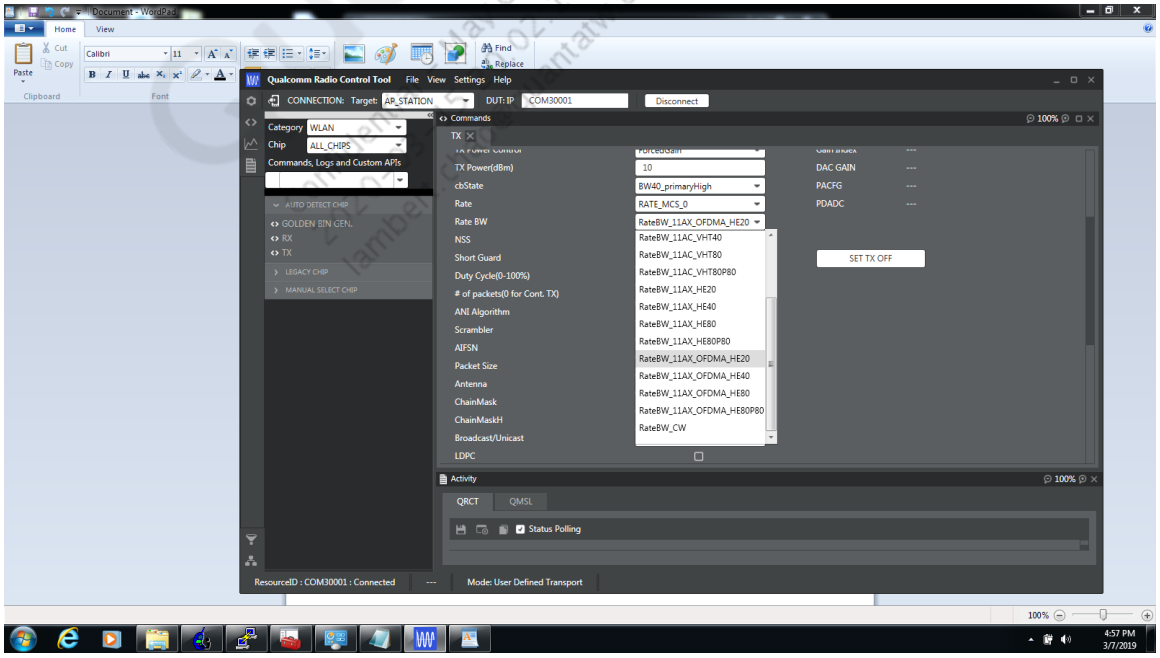


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2. Click in the designated area to send the selected tone file to DUT.

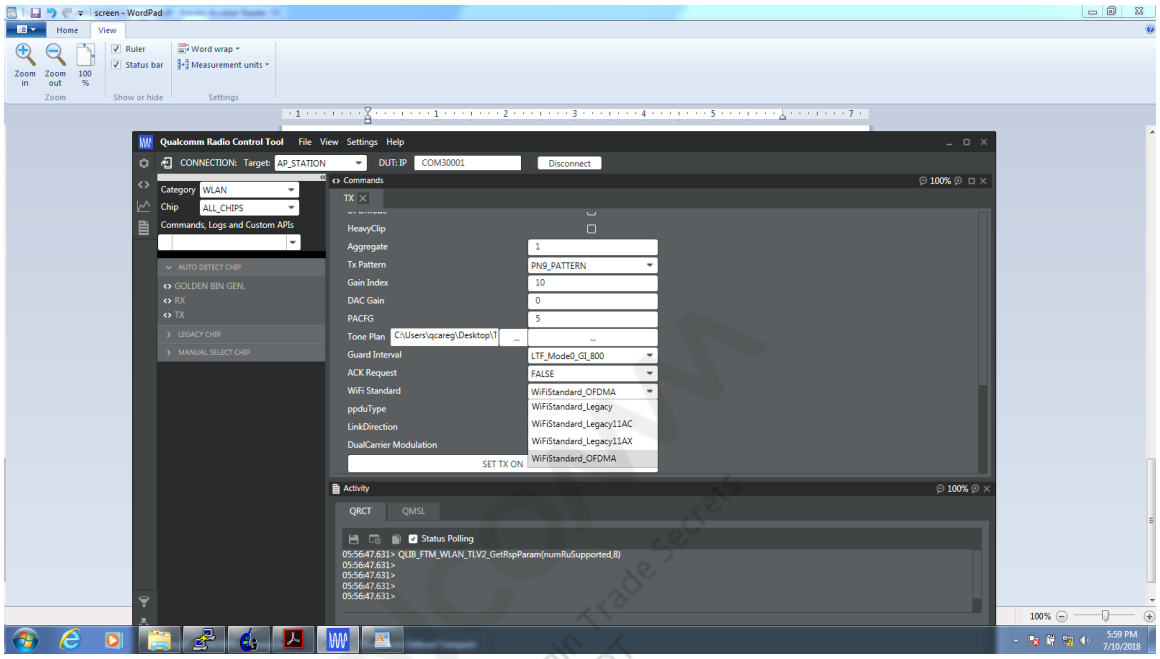



3. In the Rate BW field, select the appropriate **test tone bandwidth** from the dropdown menu.




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4. In WiFi Standard field, select **WiFiStandard_OFDMA** for any of the 4.1-4.3 MU tones.




5. Click SET TX ON to start transmission.

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# 5 Conformance Test Guidance for UL OFDMA (Client Use Case)

NOTE: This document is for use by the conformance test lab.
The information in this chapter must not be provided to the FCC and must not be included in the formal test reports.

UL OFDMA packets transmitted by a Client device will use 20, 40 or 80 MHz packet bandwidth and applicable HE_20, HE_40 or HE_80 tone maps per diagram below.

Per the diagram, these UL OFDMA packets can be transmitted using either 20 or 40 MHz legacy preambles. The 40 MHz preamble is only transmitted when the AP assigns one of the RU sizes and positions marked in red. The 40 MHz preamble is required per the 802.11ax standard because those particular positions in the channel overlap two legacy 20 MHz segments.

**Table 5-1 UL OFDMA 20 vs. 40 MHz preamble cases, Client chips only**

| All use Legacy Preamble with larger BW than the assigned RU) | | | | | |
|---|---|---|---|---|---|
| | LEGACY PREAMBLE (MHz) | RU26 | RU52 | RU106 | RU242 |
| 2.4GHz | 20 | Supported | Supported | Supported | Supported |
| | 40 | Not supported | Not supported | Not supported | Not supported |
| 5GHz | 20 | Supported | Supported | Supported | Supported |
| | 40 | Supported | Supported | Supported | Supported |

Note the UL OFDMA CTL Groups share the same compliant powers for 20 and 40 MHz preamble cases shown above. And it is known from prior validation testing that the 40 MHz preamble case results in lower compliant power levels due to band edge limits.

Therefore, conformance testing and CTL data collection must take above into account. However, conformance testing for 40 MHz preamble case must follow the 802.11ax standard such that only the channelization and tone maps described below are valid cases where the 40 MHz preamble should be used for conformance testing (note these cases are separate from UL OFDMA packets using one full bandwidth RU. In those cases, the legacy preamble is the same bandwidth as the RU):


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# 5.1 Valid tone maps and RU positions for generating UL OFDMA packets with 40 MHz preamble (instead of 20 MHz preamble)

- Valid band: 5 GHz only
- Valid Client device capability bandwidth: 80 MHz
- Valid packet type: UL OFDMA
- Valid RU sizes: RU26, 52, 106 or 242
- Valid Tone_Map used by Client: HE_80 (per below)

Valid index positions used by Client: Only Red index positions shown in HE80 tone map below:

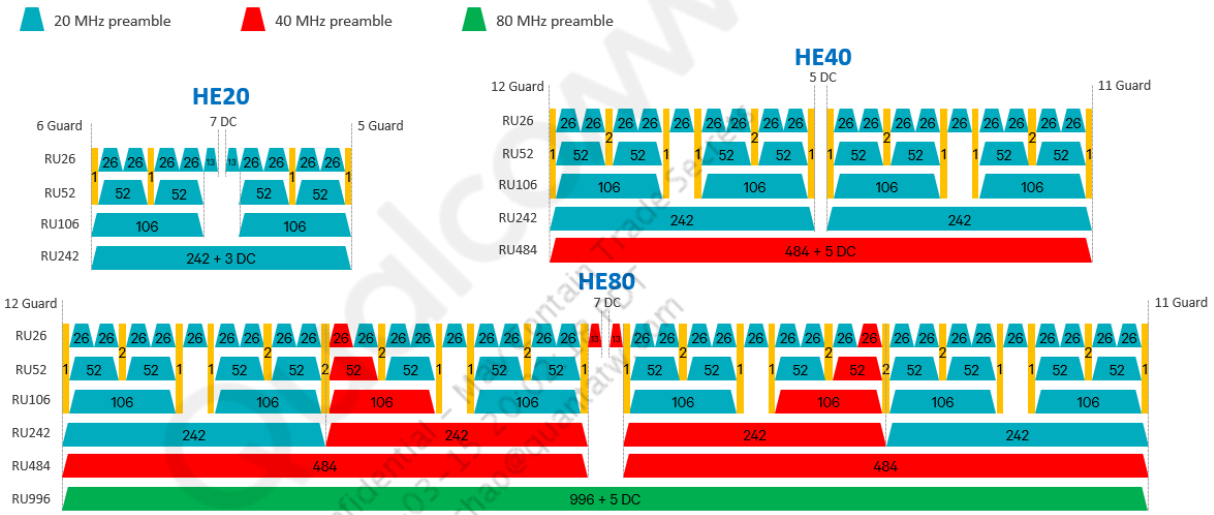
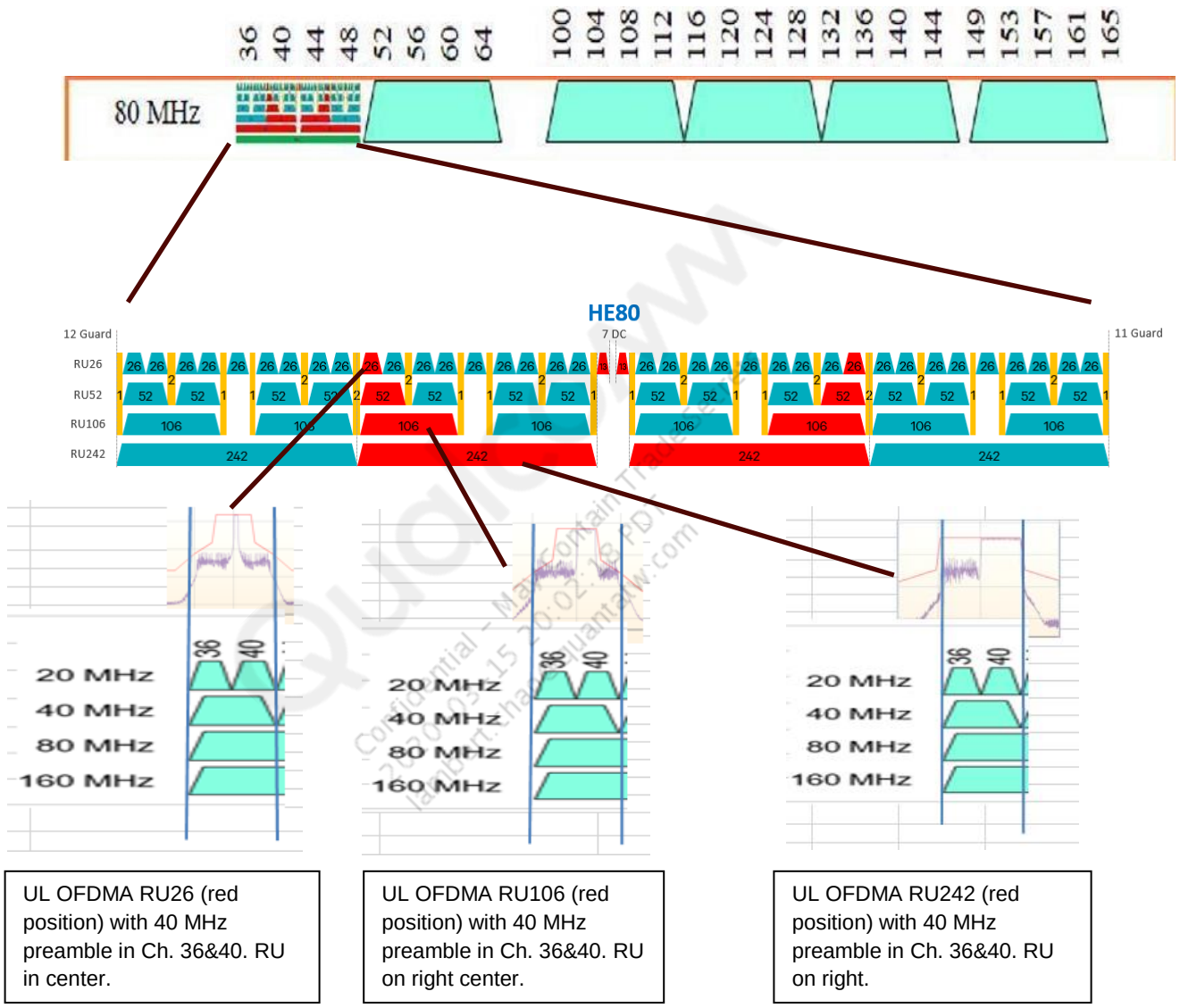



**Figure 5-1 Legacy preambles transmitted with UL OFDMA RUs**

**Test Guidance in Summary:** Conformance testing needs to be performed and CTL data collected for UL OFDM RU26, RU52, RU106 and RU242 using **40 MHz preambles** to capture the worst case compliant power. Qualcomm provides sample tone files and instructions to generate these packet types in TX99 mode.


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The following diagram illustrates valid 40 MHz preamble index positions and resulting frequency placement due to the overlap of HE_80 tone map with 802.11 channels.




UL OFDMA RU26 (red position) with 40 MHz preamble in Ch. 36&40. RU in center.

UL OFDMA RU106 (red position) with 40 MHz preamble in Ch. 36&40. RU on right center.

UL OFDMA RU242 (red position) with 40 MHz preamble in Ch. 36&40. RU on right.

**Figure 5-2 Valid 40 MHz preamble index positions and resulting frequency placement**

These examples all use the lowest 80 MHz channel placement. Upper 802.11 channels would use RU locations symmetric to those shown above (i.e., center left or fully left instead of center right and fully right as in above examples.

Finally, note the Client device is following the HE_80 tone map (assigned by the AP) but the client is transmitting an UL OFDMA packet (in these examples) using a 40 MHz legacy preamble and smaller BW RU. When assigned non-red positions by the AP, the client will do the same but transmit using 20 MHz legacy preamble centered in the applicable 802.11 20 MHz channel.


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## 5.2 Waveforms for conformance testing of UL OFDMA

To facilitate testing of UL OFDMA waveforms for formal conformance testing, QTI supplies a set of UL OFDMA tone files for use with QRCT/QSPR.

**NOTE:** Only a subset of these files are needed for formal conformance testing. Review Chapter 0~~6~~ for guidance on which waveforms/files to use for each conformance test type.

**NOTE:** Test tone XML files are available for the WLAN client device software release packages. The file is available at "/bdfUtil".

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## Table 5-2 UL OFDMA waveforms for conformance testing in FTM mode (Tx99)

| Tone Plan | Packet Length | Tone Map / Tone XML File Used with QRCT (Blue RU is modulated) | MCS | LTF GI | Preamble BW | Packet BW and Tone Map in QRCT | RU Size | RU Location (L, M, H) |
|---|---|---|---|---|---|---|---|---|
| 10-S | Short | b00000000__26_26_26_26_26_26_26_26_26 | 9 | 2x 1.6 | 20 MHz | HE20 | 26 | Low: Position 1 of 9 |
| 10-L | Long | | | 4x 3.2 | | | | |
| 20-L | Long | b00000000__26_26_26_26_26_26_26_26_26 | 9 | 4x 3.2 | | | 26 | Mid: Position 5 of 9 |
| 30-L | Long | b00000000__26_26_26_26_26_26_26_26_26 | 11 | 4x 3.2 | | | 26 | High: Position 9 of 9 |
| 50-L | Long | b00001111__52_52_26_52_52 | 6 | 4x 3.2 | | | 52 | Low: Position 1 of 4 |
| 60-L | Long | b00001111__52_52_26_52_52 | 9 | 4x 3.2 | | | 52 | Mid: Position 2 of 4 |
| 70-S | Short | b00001111__52_52_26_52_52 | 11 | 2x 1.6 | | | 52 | High: Position 4 of 4 |
| 70-L | Long | | 11 | 4x 3.2 | | | 52 | High: Position 4 of 4 |
| 80-L | Long | b0110y1y0z1z0__106_106 | 4 | 4x 3.2 | | | 106 | Low: Position 1 |
| 90-L | Long | b0110y1y0z1z0__106_106 | 1 | 4x 3.2 | | | 106 | High: Position 2 |
| 95-L | Long | b11000y2y1y0__242 | 4 | 4x 3.2 | | | 242 | Preamble and Data both 20MHz |

| Tone Plan | Packet Length | Tone Map / Tone XML File Used with QRCT (Blue RU is modulated) | MCS | LTF GI | Preamble BW | Packet BW and Tone Map in QRCT | RU Size | RU Location (L, M, H) |
|---|---|---|---|---|---|---|---|---|
| 100-L | Long | First 20MHz segment:<br>b00000000__26_26_26_26_26_26_26_26_26<br>Second 20MHz segment:<br>b00000000__26_26_26_26_26_26_26_26_26 | 9 | 4x 3.2 | 40 MHz | HE40 | 26 | "Red" position (10th of 37) |
| 105-L | Long | First 20MHz segment:<br>b00000000__26_26_26_26_26_26_26_26_26<br>Second 20MHz segment:<br>b00000000__26_26_26_26_26_26_26_26_26 | 9 | 4x 3.2 | | | 26 | "Red" position (28 of 37) |
| 110-S | Short | First 20MHz segment: b01110000__52_52_52_52 | 6 | 2x 1.6 | | | 52 | "Red" position (5th of 16) |
| 110-L | Long | Second 20MHz segment: b01110000__52_52_52_52 | | 4x 3.2 | | | | |
| 120-L | Long | First 20MHz segment: b0110y1y0z1z0__106_106<br>Second 20MHz segment: b0110y1y0z1z0__106_106 | 4 | 4x 3.2 | | | 106 | "Red" position (3rd of 8) |
| 125-L | Long | First 20MHz segment: b0110y1y0z1z0__106_106<br>Second 20MHz segment: b0110y1y0z1z0__106_106 | 4 | 4x 3.2 | | | 106 | "Red" position (6th of 8) |
| 130-L | Long | First 20MHz segment: b11000y2y1y0__242<br>Second 20MHz segment: b11000y2y1y0__242 | 4 | 4x 3.2 | | | 242 | Low position in 40MHz |
| 135-L | Long | First 20MHz segment: b11000y2y1y0__242<br>Second 20MHz segment: b11000y2y1y0__242 | 4 | 4x 3.2 | | | 242 | High position in 40MHz |
| 150-L | Long | b11001y2y1y0__484 | 4 | 4x 3.2 | 40 MHz | HE40 | 484 | Preamble and Data both 40MHz |
| 160-L | Long | b11010y2y1y0__996 | 4 | 4x 3.2 | 80 MHz | HE80 | 996 | Preamble and Data both 80MHz |

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# 6 Conformance Operating Modes for UL OFDMA (Client Use Case)

**Table 6-1 11x Client device use case – Worst case modes for FCC conformance testing**

| | 5 GHz In-Band PSD | 2 GHz In-Band PSD |
|---|---|---|
| **Small Bandwidth UL OFDMA (RU26, RU52, RU106)** | At the same power setting, RU26 will produce higher peak PSD compared to RU52 and RU106. Therefore, the Qualcomm design uses separate CTL groups for RU26, RU52 and RU106. In 5 GHz, compliant power for RU52 will be ~3 dB higher than RU26. Power for RU106 will be ~3 dB higher than RU52. | At the same power setting, RU26 will produce higher peak PSD compared to RU52 and RU106. Therefore, the Qualcomm design uses separate CTL groups for RU26, RU52 and RU106. FCC compliant powers will be similar for these RUs (since not PSD-limited). ETSI and other regions will result in compliant power differences between RU sizes as observed for 5 GHz case. |
| **Full Bandwidth UL OFDMA (RU242, RU484, RU996) vs. Single User (HE20, HE40, HE80)** | Separate power settings (i.ee ,CTL Groups) are used for RU242 vs. HE20 due to Band Edge compliance differences. Same for 40 and 80 MHz bandwidths. | Same guidance as for 5 GHz. |
| **20 vs. 40 MHz Legacy Preamble with RU26, RU52, RU106, RU242** | These CTL groups share the same power settings for 20 and 40 MHz preamble waveforms. FCC PSD compliance is the same regardless of preamble (20 vs. 40). However, see BE explanation for compliant power reduction for 40 MHz preamble waveforms. Also, Japan and Korea PSD limits reduce the compliant power for 40 MHz preamble waveforms. | 40 MHz Preamble is not used with RU26/52/106/242 per the 802.11 standard. |
| **RU Position (i.e., index) within the Channel** | There is no significant difference in achievable compliant power due to varying the position of the UL OFDMA RU within the WLAN channel. The CTL power setttings are the same for each RU position for a given RU Bandwidth. Note that 6 dB and 99% BW measurement may be smaller for RU26 and RU52 in the middle index positions. However, this does not impact resulting in-band Power and PSD results. | Same guidance as for 5 GHz. |
| **Packet Length of UL OFDMA waveforms** | There is no significant difference in achievable compliant power for UL OFDMA packets configured with very short or very long data payloads. | Long packets were observed to result in slightly lower in-band compliant power (limited by peak PSD test). Therefore, Long packet types should be used for all 2.4 and 5 GHz testing. |
| **MCS and Guard Interval for UL OFDMA RU26, RU52, RU106, RU242, RU484, RU996** | There is no impact to conformance results based on choice of Guard Interval value. Generally, the lowest MCS value should be used for In-band conformance testing since the highest target power is configured by the chip for the lowest MCS. | Same guidance as for 5 GHz. |

| | 5 GHz Band Edge Testing | 2 GHz Band Edge Testing |
|---|---|---|
| **Small Bandwidth UL OFDMA (RU26, RU52, RU106)** | At the same power setting, similar compliance margins are observed for RU26, RU52 and RU106. However, since these RUs use different power settings, BE must be tested for each RU size. | At the same power setting, larger RUs produce slightly lower BE compliance margin. FCC BE testing may use RU106 as the worse case. For other regions, these RUs use different power settings, therefore, BE must be tested for each RU size. |
| **Full Bandwidth UL OFDMA (RU242, RU484, RU996) vs. Single User (HE20, HE40, HE80)** | Compliant power due to BE conformance may be significantly lower for RU242 compared to HE20 (same for RU484 and RU996). This is due to high PAPR of legacy preamble transmitted for OFDMA. This difference may vary depending on the heavy clip values and the chip generation. Therefore, BE conformance testing should be performed for both OFDMA (RU) and Single User (HE) waveforms. | Same guidance as for 5 GHz Band Edge Testing. |
| **20 vs. 40 MHz Legacy Preamble with RU26, RU52, RU106, RU242** | Compliant power due to BE conformance may be significantly lower for RU26 with 40 MHz preamble compared to 20 MHz preamble. Same is true for RU52/106/242. This is due to the wider 40 MHz waveform in BE channels. This difference may vary depending on the chip generation. Therefore, BE conformance testing should be performed using 40 MHz preambles for BE channels. See additional test guidance notes in this bulletin since 40 MHz preambles are transmitted using a limited number of RU configurations. | 40 MHz preamble No Applicable in 2.4 GHz |
| **RU Position (i.e., index) within the Channel** | No difference in BE compliance was observed based on choice of RU position. BE testing of UL OFDMA may be performed using any one RU index. | Same observation as for 5 GHz Band Edge compliance. |
| **Packet Length of UL OFDMA waveforms** | There was no observed difference in BE compliance margin for short vs. long packet types. | There was no observed difference in BE compliance margin for short vs. long packet types. |
| **MCS and Guard Interval for UL OFDMA RU26, RU52, RU106, RU242, RU484, RU996** | There is no impact to BE conformance margin based on choice of Guard Interval value. Generally, the lowest MCS value should be used for BE conformance testing since the highest target power is configured by the chip for the lowest MCS. | Same guidance as for 5 GHz. |

| | 2 GHz and 5 GHz Radiated Spurious Emissions |
|---|---|
| **Small Bandwidth UL OFDMA (RU26, RU52, RU106)** | The smallest RU size (RU26) results in lowest RSE compliance margin. Since RU26, 52, 106 use different power settings, RSE should be checked separately for each RU size. |
| **Full Bandwidth UL OFDMA (RU242, RU484, RU996) vs. Single User (HE20, HE40, HE80)** | RSE should be performed for each Single User waveform (HE20, HE40, HE80). Although power settings for RU242, RU484, RU996 are generally lower than HE20, HE40, HE80 respectively, RSE should be spot checked for these waveforms also due to higher PAPR. |
| **20 vs. 40 MHz Legacy Preamble with RU26, RU52, RU106, RU242** | 40 MHz Preamble may result in worse spurious emissions for Band Edge channels. And 20 and 40 MHz preambles share same CTL powers. Therefore, test RSE at band edge channels using 40 MHz preamble waveforms at applicable compliant BE channel target powers. Other channels may use either 20 or 40 MHz preamble waveforms for RSE (also at the applicable in-band compliant power targets. |
| **RU Position (i.e., index) within the Channel** | No difference in RSE was observed based on choice of RU position. RSE for UL OFDMA may be performed using any one RU index. |
| **Packet Length of UL OFDMA waveforms** | There was no observed difference in RSE levels due to packet length differences. |
| **MCS and Guard Interval for UL OFDMA RU26, RU52, RU106, RU242, RU484, RU996** | There is no impact to BE conformance margin based on choice of Guard Interval value. RSE should be checked using a high and low MCS value for each waveform since spurious emissions may vary for the higher order modulation of high MCS values (even though in-band target power will be lower). |

# 7 Worst-Case Mode Justification

The text in this chapter may be used as the basis for a worst-case mode justification to share with a test lab, as part of the manufacturer's conformance test plan.

**The manufacturer must modify the text (in red) where required to fully reflect the product configuration under test.**

NOTE: Any portion of the text below may be used by the 3rd party test lab, to the extent required, for inclusion in the formal test reports.

## 7.1 Testing worst-case modes

### 7.1.1 Antenna gain

**[Case 1] One antenna configuration**

The product is configured with one antenna option. All RF conformance testing was performed using the highest antenna gain in 2.4 and 5 GHz bands.

Each antenna element is the same.

The resulting per-chain conducted powers captured from conformance testing reflect the declared antenna gains for each band per the provided antenna spec sheets.

**[Case 2] Multiple antenna options sharing the same conducted powers**

The product supports multiple antenna options (the antenna elements used for each chain are always the same). All RF conformance testing was performed using the highest gain antenna option for each band. The resulting per-chain conducted powers captured from conformance testing reflect the declared antenna gains for those highest gain antennas. The end-product will use the same per-chain powers for the other available antenna options with lower gain.

**[Case 3] Multiple antenna options and collection of multiple sets of compliant powers**

The product supports multiple antenna options (the antenna elements used for each chain are always the same).

The RF conformance testing was repeated using antennas with different peak gains. Multiple sets of conducted per-chain compliant powers were captured during conformance testing. Separate product SKUs will be created using the applicable set of compliant powers for the antenna option included with the product. There is no facility for the end-product to implement multiple sets of compliant powers in the same product SKU (i.e., power is not dynamically adjusted based on antenna option).

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### 7.1.2 CTL groups

By design, each entry/row within a CTL group uses *the same compliant power* values, across all modulations and operating modes for that group.

The same compliant power is shared for all operating modes within a CTL group, The test plan reflects validation and formal conformance testing to determine the worst case compliant power covering all modes used in the CTL Group, Full details of worst case validation evaluation are detailed in this document.

During conformance testing, a single set of compliant powers are applied to every modulation and operating mode contained in CTL group. The worst-case modulation/operating mode is defined as the mode within a CTL group that creates the highest in-band and out-of-band emissions during conformance testing and which is used to perform the formal conformance testing..

### 7.1.3 Worst-case modulation/MCS within a CTL group

The lowest data rate within each CTL group will result in the highest in-band transmitted power and PSD and therefore is considered the Worst-Case mode. Board calibration power for those lowest rates are set higher than for the higher rates. Therefore, conformance testing using the lowest data rate captures the highest transmit power for the CTL group. Testing is also performed using higher MCS/rates for Out-of-Band emissions.

### 7.1.4 Short and long guard interval

Short and long guard interval (GI) settings are invoked by software during WLAN operation. All solutions use cyclic prefix to completely fill the guard interval, regardless of the GI value. There is no measurable difference in the resulting transmitted waveforms when the GI value changes. For convenience, the shortest GI is used for all conformance testing. There is no justification to repeat any conformance testing using alternate GI values.

### 7.1.5 Beamforming

Beamforming may be enabled for various 802.11 modulations/modes. The CTL groups in the test plan indicate if beamforming is enabled for one or more modes. Different peak power and PSD regulatory limits apply for those modes. Compliant powers, i.e., CTL data, will be collected for each beamforming mode in the CTL groups indicated in the test plan.

The conformance test software does not specifically enable beamforming during conformance testing (i.e., using QRCT test software[1]). This is because the beamforming feature simply adjusts phase of transmit chains based on higher layer protocols. This behavior has no impact on conducted emissions measurements. However, the test lab applies different FCC and ETSI limits for the beamforming modes within a CTL group.

The compliant powers captured from conformance testing for each beamforming and non-beamforming mode are included in the final CTL Tables programmed in the product during manufacturing.

[1] For this reason, QRCT test software includes no option/setting to enable beamforming.


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### 7.1.6 MIMO spatial streams

FCC and ETSI rules impose different transmit power limits depending on the number of spatial streams (Nss). A single design (especially an access point) will support transmission of many combinations of spatial streams. The QTI CTL solution captures separate CTL data (based on conformance results) for each transmitted spatial stream. The same CTL data may be shared or configured differently for each combination of spatial stream. The provided test plan indicates for each CTL group, the desired target compliant power for each stream combination. The test lab must apply the correct peak power and PSD limit for each MIMO mode and compile compliant data for each CTL group. The applicant will provide detailed guidance to the lab including desired target powers and the array gain calculations applicable to each MIMO mode.

The compliant powers captured from conformance testing for each MIMO mode are included in the final CTL Tables programmed in the product during manufacturing.

### 7.1.7 Fall back to fewer number of Tx chains

The QTI WLAN solution supports automatic fallback to transmit some packets using a subset of the total number of transmit chains present in the hardware. This behavior is invoked dynamically by the software based on use case and link conditions.

[Manufacturer must delete the section which does not apply.]

**Default CTL behavior**

Same per-chain power regardless of number of active chains:

The product uses the same per-chain CTL value regardless of number of active chains. The total power will therefore be lower when fewer chains are active. Conformance testing for case of fewer active transmit chains is not required.

OR

**The product supports different per-chain power, i.e., higher, when fewer chains are active**

Additional conformance testing is to be performed to verify compliance and determine CTL values for these fallback modes.

The provided test plan indicates the fall back modes supported along with target per-chain compliant powers for each supported fall back mode.

The compliant powers captured from conformance testing for each separate fall back mode will be included in the final CTL Tables programmed in the product during manufacturing.

### 7.1.8 CBW configuration during conformance testing using QTI FTM and QDART

IEEE 802.11n/ac products support wide channel bandwidths, i.e., 40, 80, 160 MHz) but transmit packets using various smaller bandwidths, e.g., 20 MHz, during operation.

- Capability bandwidth (CBW) is used for the max bandwidth the hardware is capable of transmitting. CBW is generally a static characteristic set in the factory and determined by the chip and radio design, e.g., an 802.11ac device with a CBW of 80 MHz.
- Packet bandwidth (PBW) is used for the bandwidth of the currently transmitted packet, i.e., HT20 or VHT80 packets.


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A reduction in CBW results in modified analog filtering and DAC/ADC behavior which may affect in band power/PSD and out of band emissions conformance results. *The worst-case conformance results are obtained using the widest supported CBW setting*. Also note: The QTI CTL scheme uses the same CTL values for all CBW settings.

**Therefore, the widest CBW must be used for conformance testing of all packet types so the final CTL data reflects the worst case for all supported CBW options.**

Instructions for using test software are provided in Chapter 10, indicating how to ensure the widest supported CBW is used during conformance testing.

### 7.1.9 DL OFDMA Packets (Access Point Only)

[Manufacturer must delete the section which does not apply.]

Qualcomm 802.11ax Access Points ensure DL OFDMA packets assign and modulate 100% of the RUs. Partially populated DL OFDMA packets are not supported. Therefore, testing of this packet type uses fully modulated packets. The partially modulated waveforms per FCC guidance is not applicable for this product.

OR

Qualcomm 802.11ax Access Points ensure DL OFDMA packets assign and modulate at least 50% of the RUs. The chipset will back off the per-tone power for 'partially modulated' packets so the worst case PSD remains similar to the fully modulated case.

Since either the fully or partially modulated cases may result in worst case PSD and band edge compliance, the FCC guidance for finding the worst case values was followed. See Chapter 4.

Qualcomm provides DL OFDMA waveforms for conformance testing per FCC guidance:

- Fully modulated
- Partially modulated – at edges of the channel
- Partially modulated in center of the channel

### 7.1.10 UL OFDMA Packets (Client Device Only)

802.11ax Client devices transmit 802.11ax single user (HE_SU), UL MU-MIMO and UL OFDMA packet types.

UL OFDMA packets are comprised of a full bandwidth legacy preamble (20, and in some cases, 40 MHz) followed by one assigned RU bandwidth. Therefore, the bandwidth varies during transmission of a single packet under test. Also the length of the data portion of the packets will vary depending on the queued data and link characteristics. Also multiple MCS and Guard Interval values may be used for each packet.

Extensive validation testing was performed while varying the parameters above to determine the worst case conformance results (i.e., in-band power, PSD, Band Edge levels and Radiated Spurious Emissions. See Chapter 5 for full details and test guidance.


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## Guidance from Prior Worst Case Mode Validation Testing of UL OFDMA Waveforms

Guidance below applies to both 2.4 and 5 GHz bands unless otherwise noted.

### Small RUs: UL OFDMA RU26, RU52, RU106, RU242, RU484, RU996

Compliant powers must be determined for UL OFDMA waveforms. Separate CTL tables are populated for each. RU26 will yield highest PSD result for bands and regions that are PSD-limited.

Band Edge results were very similar for each RU size. Therefore band edge testing and Radiated Spurious Emissions should be performed using the one RU size with the highest expected in-band power.

However, Radiated Spurious Emissions should also be checked using RU26 which resulted in a smaller margin of compliance.

### RU Position Within the Channel (small RU cases): RU26, RU52, RU106

The RU position within the channel (low, mid vs. high) did not impact any of the conformance results. Any position (i.e., index) RU may be chosen for formal conformance testing.

### Full Bandwidth UL OFDMA 1 RU vs. HE_SU_ Waveforms

UL OFDMA using a single full BW RU is supported by Qualcomm Client chipsets (e.g.. OFDMA_20 with RU242, OFDMA_40 with RU484, OFDMA_80 with RU996). Conformance testing is required separately for these waveforms vs. HE_SU_20, 40, 80 since the EVM requirements for these two sets of waveforms are different. Therefore, PSD and BE results may be different. Separate CTL Groups are used for UL OFDMA (full BW) vs. HE_SU. Also note HE_SU waveforms share the same CTL Groups as legacy HT/VHT waveforms.

#### UL MU-MIMO Waveforms

Future Client chipsets will add 11ax UL MU_MIMO_20, 40 and 80 waveforms. These will share the same CTL groups as HT/VHT/HE_SU waveforms. MU_MIMO waveforms are comprised of the same modulation scheme, data and tone maps as HE_SU waveforms. MU_MIMO packets only differ in the transmitted spatial dimension and are not expected to demonstrate measurable difference in compliance results compared to HE_SU waveforms.

Qualcomm Access Point solutions currently support DL MU-MIMO waveforms. Per above, these share the same CTL Group as HT/VHT/HE_SU waveforms.

### 40 MHz Legacy Preamble with UL OFDMA RU26, RU52, RU106 and RU242 in 5 GHz

The 802.11 standard requires the above UL OFDMA RU configurations to support transmission of both 20 and 40 MHz legacy preambles. The 40 MHz preamble is transmitted for a subset of RU positions in the 802.11ax tone maps.

Compliant power for BE channels will be lower for UL OFDMA packets with a 40 MHz preamble compared to a 20 MHz preamble. The applicable waveforms with 40 MHz preambles should be used for determining in-band power targets for all BE channels. See Chapter 3 for valid UL OFDMA packets types that will transmit 40 MHz preambles.

Note that choice of 20 vs. 40 MHz preamble for RU26 to RU242 does not impact other compliance results (i.e., in-band PSD or RSE).


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### Short and Long Packet Lengths

Validation testing showed that packet length (shortest vs. longest theoretical packet lengths supported by the 802.11ax spec and Qualcomm chipsets) had no impact on BE and RSE conformance results. Long packet types did result in slightly higher PSD results. Also, long packet lengths generate more consistent high duty cycle operation (i.e.,TX99 mode) during conformance testing.

Therefore, long packet types should be used for all conformance testing.

### MCS and Guard Interval (GI) for UL OFDMA packets

Validation testing showed there is no impact to conformance results for different GI values.

As is true for legacy waveforms, for UL OFDMA, the lowest MCS value should be used for In-band conformance testing since the highest target power is configured by the chip for the lowest MCS.

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# 8 Conformance Test Plan/Checklist

The test lab must receive a conformance test plan from the manufacturer/applicant identifying the supported operating modes. Manufacturers may customize MIMO behavior, beamforming and fall back operation using fewer transmit chains. These product-specific decisions are identified in the conformance test plan.

The test plan should include the manufacturer's *target conducted transmit powers for each operating mode*. The lab should not attempt to prove compliance using powers any higher than the targets. However, it is common to find that compliance for various modes (especially band edge channels) can only be achieved at levels lower than the targets provided by the manufacturer.

Conformance testing takes into account the specific antennas chosen by the manufacturer for the end-product. The test plan and CTL strategy must take into account the case of multiple antenna options included in the scope of the conformance testing.

Conformance testing of every combination of supported modulation, bandwidth and MIMO mode is not feasible for WLAN devices. Regulators accept conformance test reports that identify "worst-case" modes for testing if a justification is provided.

The conformance test plan indicates that only one modulation/modes within each CTL group needs to undergo conformance testing for in-band peak power and PSD tests). The justification text of worst-case modes is found in Chapter 7.

NOTE: It is highly recommended that the manufacturer include a preliminary CTL file to the test lab at the start of the conformance test project. The CTL file contains all target powers for each supported operating mode. Any portion of the text below may be used by the 3rd party test lab, to the extent required, for inclusion in the formal test reports.

## 8.1 Supported operating modes checklist

The manufacturer may use the following checklist as part of the test plan provided to the test lab to indicate all operating modes supported by the product.

This document is focused on RF power and emissions conformance testing. However, items are included in the checklist to assist in the test planning for the non-RF testing which may also be required. The non-RF tests are out of the scope of this document (i.e., DFS, radar detection).


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## Product Capabilities/Features

[Entries expected to be supported by all current chipsets are already checked. Manufacturers must check additional boxes to fully match the product under test.]

**a. Regulatory domains included in scope of conformance project:**

- ☐ USA [FCC]
- ☐ Canada [ISED]
- ☐ Japan [MIC/MKK]
- ☐ European Community [ETSI]
- ☐ China [SRRC]
- ☐ South Korea
- ☐ Taiwan
- ☐ Australia and New Zealand
- ☐ _____
- ☐ _____
- ☐ _____
- ☐ _____

**b. 5 GHz channel bandwidths:**

- ☒ 20 MHz
- ☒ 40 MHz
- ☒ 80 MHz
- ☐ 160 MHz (contiguous)
- ☐ 80+80 MHz (non-contiguous)

**c. 2.4 GHz channel bandwidths:**

- ☒ 20 MHz
- ☒ 40 MHz

**d. Supported channels:**

- ☒ Supports All Standard 802.11 center frequencies for each Bandwidth
- ☒ For U.S., supports U-NII2-3 overlap channels [center frequencies 5720 (20 MHz), 5710 (40 MHz), 5690 (80 MHz)]

**e. Number of Transmit/Receive Chains Present in the Hardware**

- ☐ 8x8
- ☐ 4x4
- ☐ 2x2
- ☐ 1x1

**f. For fall back operation (using fewer than the total number of transmit chains), will the device use higher-per chain transmit power (compared to operation using all chains)?**

- ☐ No (Skip to g.)
- ☐ Yes (Check each supported case)
  - ☐ Higher per-chain power when falling back to 4 Chain operation
  - ☐ Higher per-chain power when falling back to 2 Chain operation
  - ☐ Higher per-chain power when falling back to 1 Chain operation

NOTE: The product will always use the same per-chain power for the currently active chains.

**g. Multiple antenna options:**

- ☐ **Only one antenna option is implemented/offered for sale with device**
- ☐ More than one **antenna option is implemented/offered for sale with device**

[spec sheets provided]

**h. 5 Ghz antennas:**

- ☐ Identical 5 GHz antenna elements are used on all Tx/Rx chains
- ☐ Different 5 GHz antenna elements are used on different transmit/Tx chains [Manufacturer must provide specs to test lab]

**i. 2.4 Ghz antennas:**

- ☐ Identical 2.4 GHz antenna elements are used on all Tx/Rx chains
- ☐ Different 2.4 GHz antenna elements are used on different transmit/Tx chains [Manufacturer must provide specs to test lab]

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| **Product Capabilities/Features**<br>[Entries expected to be supported by all current chipsets are already checked. Manufacturers must check additional boxes to fully match the product under test.] | |
|---|---|
| **j.** | **Beamforming and MIMO modes:**<br>☒ Beamforming (all current 802.11ac and later designs support beamforming.<br>The specific operating modes supporting beamforming are identified in the preliminary CTL Table provided by the manufacturer.<br>☒ MIMO modes (multiple spatial streams)<br>The specific operating modes supporting multiple spatial streams and their associated target transmit powers are identified in the preliminary CTL Table provided by the manufacturer. |
| **k.** | **TPC feature implemented?**<br>☒ Yes. A user interface option is available for the end-user to manually reduce transmit power through a range of 6 dB.<br>☐ Yes. The device supports 802.11 standards for Client device operation where the infrastructure equipment is capable to instruct the device under test to reduce transmit power through a range of 6 dB.<br>☐ No |
| **l.** | **DFS related operating modes (check all that apply):**<br>☐ DFS Master (implements radar detection in radar channels in order to transmit beacons)[2]<br>☐ Client with radar detection (including WDS Repeater mode)<br>☐ Client without radar detection |
| **m.** | **Additional support by Client devices (check all that apply):**<br>☐ Ad hoc operation in 5150-5250 and 5725-5850 without radar detection [TDLS, WiFi Direct, Soft-AP or proprietary].<br>☐ Ad hoc operation in 5250-5725 with radar detection [TDLS, WiFi Direct, Soft-AP or proprietary]. |
| **For ETSI RX Blocking and Adaptivity Testing** | |
| **o.** | **Load Based Equipment that supports multi-channel operation:**<br>☐ The LBE equipment supports Option 1 as described in clause 4.2.7.3.2.3<br>☒ The LBE equipment supports Option 2 as described in clause 4.2.7.3.2.3<br>▪ The (maximum) number of Adjacent channels used for multi-channel operation:<br>☐ VHT80 with 4 options for primary HT20 channel<br>☐ VHT160 with 8 options for primary HT20 channel |
| **p.** | **Adaptivity (channel access mechanism):**<br>☒ Load Based Equipment |

[2] The Radar Band is 5250-5725 MHz comprised of IEEE 20 MHz channels 52-144 (and larger Bandwidth channels in this same range). Newer designs may also support radar detection per European requirements in 5725-5850 MHz.


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## Product Capabilities/Features

[Entries expected to be supported by all current chipsets are already checked. Manufacturers must check additional boxes to fully match the product under test.]

**q.** **With regards to adaptivity for Load Based Equipment:**

☐ The Load Based Equipment operates as a *Supervising* device
Check above box if UUT is an access point

☐ The Load Based Equipment operates as a *Supervised* device
Check above box if UUT is a Client device

☐ The Load Based Equipment can operate as a *Supervising* and as a *Supervised* device
Check above box if UUT supports both AP and STA. (i.e., Hotspot, P2P GO)

The Priority Classes implemented by the Load Based Equipment:

- When operating as a *Supervising* device
  ☒ Priority Class 2
- When operating as a *Supervised* device
  ☒ Priority Class 2

☒ The Load Based Equipment can operate as an *Initiating* device and as a *Responding* device

With regard to Energy Detection Threshold, the Load Based Equipment has implemented either Option 1 of clause 4.2.7.3.2.5 of ETSI EN 301 893 V2.1.1 or Option 2 of clause 4.2.7.3.2.5 of ETSI EN 301 893 V2.1.1:

☒ Option 1
☐ Option 2
Specify which protocol has been implemented: ☒ IEEE 802.11

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# 9 Conformance Test Flow

NOTE: This document is for use by the conformance test lab.
The information in this chapter must not be provided to the FCC and must not be included in the formal test reports.

This chapter presents a suggested test flow for test lab FCC Part 15 conformance testing.

The test flow is especially important for a multi-transmitter access point product with 4 or 8 transmit chains. Such products implement many combinations of operating modes and bandwidths with many CTL Table entries with multiple target powers within each subgroup.

The test flow is meant to optimize determination of CTL values compliant with spurious/harmonics for restricted band edge, non-restricted band edge and mid-band channels.

NOTE: The following process assumes the manufacturer has provided the preliminary CTL Table to the test lab at the start of the conformance test project. It may be helpful to provide a CTL Table to the lab with *Enable In-band Flag set to 1* for easier review of the data.

## 9.1 Rationale

The preliminary CTL data (CTL spreadsheet):

- The preliminary CTL spreadsheet provided by the manufacturer contains the max. theoretically allowed in-band per-chain powers for each supported operating mode. Various entries may have also been reduced due to board limitation (i.e., calibration data) input by the manufacturer when the preliminary CTL Table was generated.
- The preliminary CTL spreadsheet does not reflect band edge channel reductions for restricted band compliance and does not reflect mid-band channel reductions for spurious emissions compliance.
- The test lab must perform compliance testing to determine which CTL Table entries need to be reduced (mid-band channel and band edge channels) to comply with spurious emissions limits.
- The test flow in Figure 9-2 dictates spurious emission testing first for a given channel and mode to determine the worst case per chain and total power. Based on that result, maximum representation for sub-cases of chain and stream power levels are applied.
- We recommend the lab uses antenna radiated out of band emission test devices with more than two transmit chains/antennas.
- For 1x1 and 2x2 client modules, conducted out of band emission test method may be used, especially for modules to be certified using multiple antenna options.


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# 9.2 Testing L/M/H channels not adjacent to restricted band

NOTE: Be aware of second harmonic falling in FCC restricted bands

## 9.2.1 Preparation

Sample CTL data for FCC 5 GHz HT/VHT20 group is shown in Figure 9-1.

| Group | Antenna used for table | Reg Domain | Beamforming | Mode | Num Chains on bit Mask | Num Streams on bit Mask | Power levels are per chain (dBm) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | FCC channel list | | | | | 5180 | 5200 | 5220 | 5240 | 5260 | 5280 | 5320 | 5500 | 5520 | 5720 | 5745 | |
| 5GHz HT20/VHT20 | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 11110000 | 00000001 | 12.0 | | | | 7.0 | | | 6.0 | | | 18.5 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 11110000 | 00000010 | 15.0 | | | | 10.0 | | | 9.0 | | | 18.5 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 11110000 | 00000100 | 17.0 | | | | 12.0 | | | 11.0 | | | 19.5 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 11110000 | 11111000 | 19.0 | | | | 14.0 | | | 13.0 | | | 20.5 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 00001100 | 00000001 | 18.0 | | | | 13.0 | | | 12.0 | | | 22.0 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 00001100 | 00000010 | 21.0 | | | | 16.0 | | | 15.0 | | | 22.0 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 00001100 | 00001100 | 22.0 | | | | 17.0 | | | | | | 22.0 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 00000010 | 00000001 | 22.0 | | | | 19.0 | | | 18.0 | | | 22.0 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | Flag |
| | | (0x10) | 0 | HT20 | 00000010 | 00000010 | 22.0 | | | | 20.0 | | | | | | 22.0 | Power |
| | | FCC | | | | | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Flag |
| | | (0x10) | 0 | HT20 | 00000001 | 00000001 | 22.0 | | | | | | | | | | | Power |

**Figure 9-1 CTL data for FCC 5 GHz HT/VHT20 group**

1. Review the CTL spreadsheet within each CTL group by bandwidth/mode.
2. Determine the different levels of power and sort from the highest to the lowest power, P1, P2, P3, … PN.
3. The result is shown in Table 9-1.

**Table 9-1 CTL spreadsheet sort results**

| n | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Pn | P1 | P2 | P3 | P4 | P5 | P6 | P7 | P8 | | |
| 5.15-5.25 GHz | 22 | 21 | 19 | 18 | 17 | 15 | 12 | | | |
| 5.25-5.35 GHz | 22 | 20 | 19 | 17 | 16 | 14 | 13 | 12 | 10 | 7 |
| 5.47-5.725 GHz | 22 | 20 | 18 | 17 | 15 | 13 | 12 | 11 | 9 | 6 |
| 5.725-5.85 GHz | 22 | 20.5 | 19.5 | 18.5 | | | | | | |

The block diagram in Figure 9-2 provides full details.


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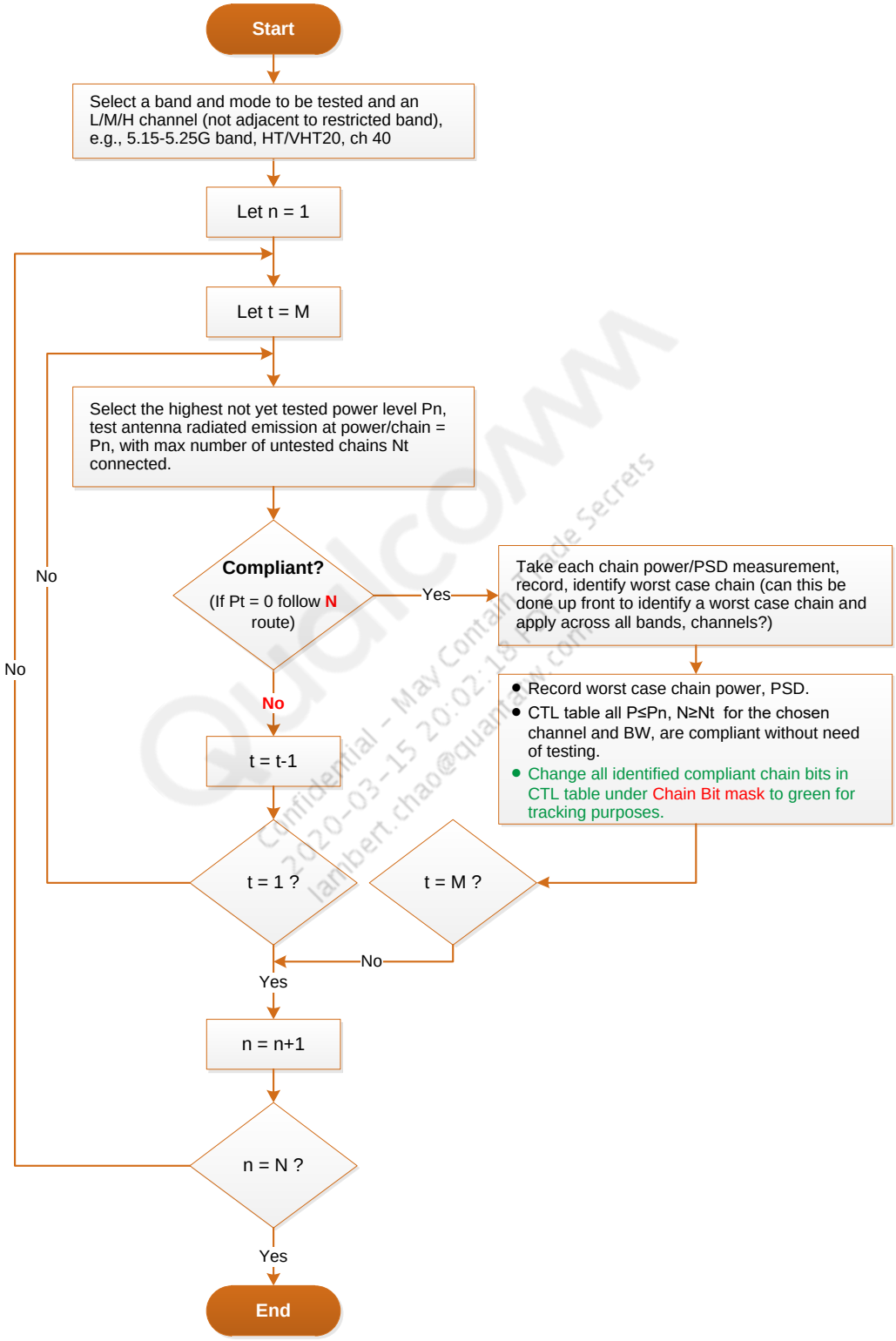



**Figure 9-2 L/M/H channels not adjacent to restricted band block diagram**

NOTE: Level of CTL power for a specific BW/mode is N, with P1 represent the higher power, PN the lowest power level and number of transmitters that may transmit simultaneously is Nt, t=1, 2,…,…, M


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### 9.2.2 Process summary

1. Test a particular mode (e.g., HT/VHT20) with antenna radiated spurious emission (all chains connect to antennas of equal gain) at per chain power =P1, with max number of chains enabled.
   a. If antenna radiated emissions pass, then test each individual chain at the same power setting for power, and PSD.
      i Identify the chain with max measured power/PSD.
      ii Test that identified chain for all other power/PSD items as representative of the worst-case chain.
      iii This prevents the need to repeat testing of the other chains.

   OR

   b. If antenna radiated emissions fail, then, reduce number of Tx chains until it passes.
      i Assume Nant=Np1, all modes of P1 limit, with Nant≤Np1 modes can pass mid band spurious emission without testing.
2. If necessary, reduce per chain power to the next level of P2.
   a. Test with all chains On, then reduce number of Tx chains till radiated spurious emission passes.
   b. Assume Nant=Np2. All Nant≤NP2 modes are compliant at equal or below P2 level power without testing.
3. Repeat the same steps for power reduction until all tests are done or covered without testing.
4. Repeat the same steps for all modes to be tested (e.g., HT/VHT40, 80, 160, etc.) and for all bands.

## 9.3 Testing band edge channels in immediate vicinity of restricted band

### 9.3.1 Process summary

1. Test BE channel channels (e.g., HT/VHT20 ch36) for radiated emission in restricted band with max number of transmitters on (Nant=8 in this example).
   a. Find passing level P(chn, Nant) (e.g., P(ch36, 8).
   b. Record min (compliance power, CTL Table power) for the channel/mode, and its corresponding PSD.
   c. Determine if the P(chn, Nant) is compliant for modes of fewer antennas (e.g., P(ch36, 8) is good for Nant=6, 7, 8, but not for Nant≤5).
   d. If not, test BE channel chn radiated emission with the next number of antennas on (e.g., Nant=5), find compliance power level P(chn, Nant) (P(ch36, 5), measure power and PSD and record. Determine if this level of power is good enough for the numbers of antennas, to further determine next level of testing.


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2. Repeat process until all tests of the same bandwidth test group are done or covered without testing for the channel.
3. Repeat with each bandwidth/mode group and band.

## Example

An example of the process using the data presented previously:

1. Attached CTL, for 5.15-5.25 GHz mid-band channel, 5220 MHz HT/VHT20 mode P1=22, P2=21, P3=19, P4=17, P5=15, P7=12

| N | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Pn | P1 | P2 | P3 | P4 | P5 | P6 | P7 |
| dBm | 22 | 21 | 19 | 18 | 17 | 15 | 12 |

2. From the CTL spreadsheet attached, max number of transmitters is 8, M=8

| t | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Nt | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

   n=1, t=8 to start with.

3. Test radiated spurious emission at channel power P1=22 dBm, with all 8 chains connected. If of passing result, the entire CTL Table 11J-31J are compliant without further testing.
4. If 1 is over limit, make t=7, test P1=22 dBm with 7 chains transmit, if radiated out of band emission is of compliant result this time, at P1=22 dB, all CTL Table 20J-31J are compliant without need of further testing. 11J-19J all Num Chains on bit Mask 01110000 are compliant without testing.
5. If still non-compliant, further reduce transmit chain numbers till pass at this power level
6. Assume if at P1=22 dBm, radiated can only pass at Nt=4, then at this point, it is clear all Nt=1, 2, 3, 4 21J-31J are compliant at it's filled in power without testing.
7. Test at P2=21 with Nt=8, for radiated emission, repeat process at P2.


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# 9.4 Testing band edge channels adjacent to restricted bands

## 9.4.1 Process summary

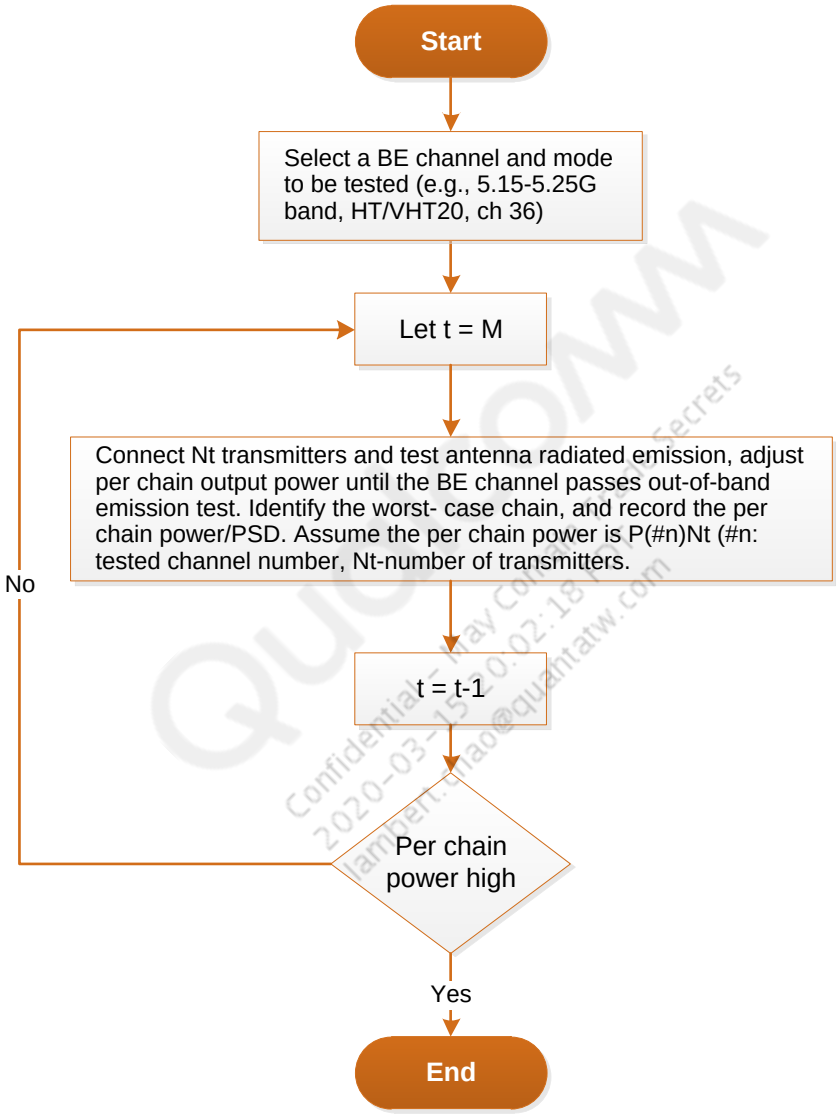



**Figure 9-3 Testing band edge channel adjacent to restricted band (simplified as BE channel)**


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# 10 Conformance Test Using QRCT

NOTE: This document is for use by the conformance test lab.
The information in this chapter must not be provided to the FCC and must not be included in the formal test reports.

Qualcomm Radio Control Toolkit (QRCT) test software is used for WLAN radio conformance testing of in-band and out-of-band emissions.

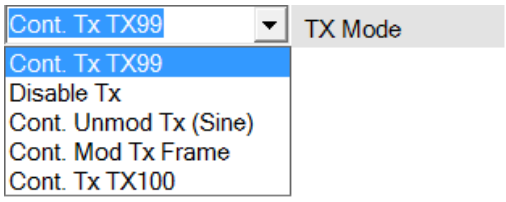
QRCT supports TX99 continuous transmission modes required per FCC and ETSI test procedures. This chapter summarizes the configuration of QRCT to perform all required FCC and ETSI RF emissions conformance testing.

QRCT is not used for "mission mode" operation where a normal WLAN link is required (such as for DFS testing).

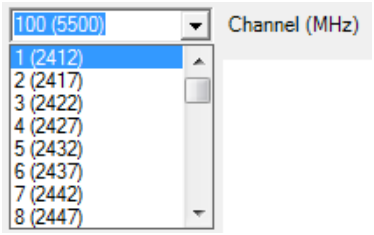
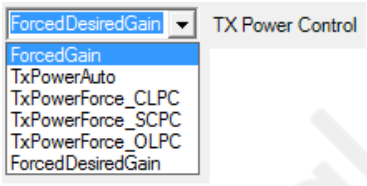
NOTE: The following configuration instructions for conformance testing also apply when using Qualcomm Sequence Profiling Resource (QSPR) test software. Although the interface differs between QRCT and QSPR, the parameter names and functions are the same.

The manufacturer/applicant must provide the necessary test sample and support equipment to operate QRCT with the device under test.

**Table 10-1 QRCT WLAN transmitter settings and testing**

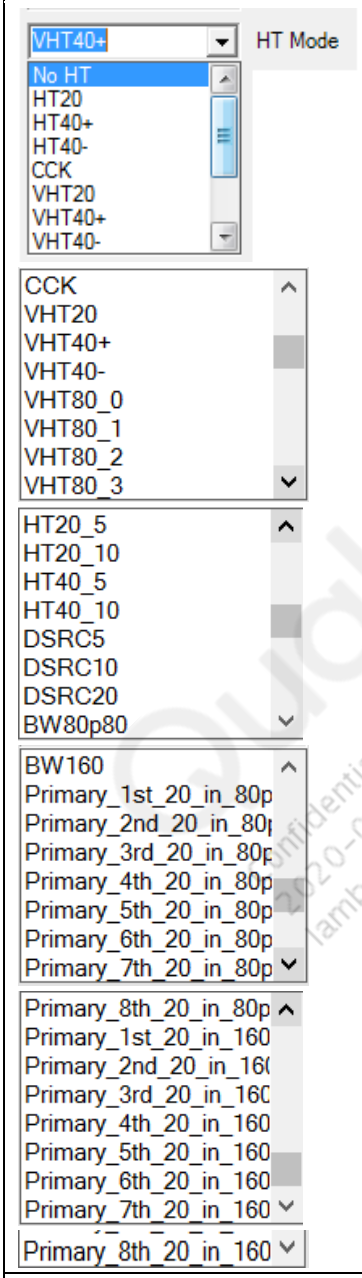
| Step | QRCT Menu Items | Options |
|---|---|---|
| 1 | Tx mode<br>  | ▪ Cont. Tx TX99: 99% duty cycle Tx<br>▪ Disable Tx: Disables Tx<br>▪ Cont. Unmod Tx (Sine): Tx CW tone<br>▪ Cont. Mod Tx Frame: Tx modulated Tx frame<br>▪ Cont. Tx TX100: 100% duty cycle Tx |
| | Use TX99 for all regulatory conformance testing of in-band transmit power and out-of-band emissions. | |


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| Step | QRCT Menu Items | Options |
|---|---|---|
| **2** | Channel<br>  | ▪ All 2G and 5G channels |
| | Select desired 2.4G or 5G channel (center frequency).<br>Use caution to choose a valid center frequency for the chosen bandwidth and modulation set in Step #5.<br>QRCT interface may allow combinations of channel and mode that may not be supported in the end product. | |
| **3** | Tx power control<br>  | ▪ ForcedGain: Not currently used<br>▪ TxPowerAuto: Auto detect power control<br>▪ TxPowerForce_CLPC: Use CLPC calibration data<br>▪ TxPowerForce_SCPC: Use SCPC calibration data<br>▪ TxPowerForce_OLPC: Use OLPC calibration data<br>▪ ForcedDesiredGain: Use fixed gain. Uses Gain Index and DAC Gain. |
| | Choose appropriate calibration type for the chip/board under test.<br>The board configuration can be read from the BDF.<br>Most current designs use CLPC (Closed Loop Power Control) | |
| **4** | Tx power<br>  | Input for Tx power in 0.5-dBm increments.<br>**<u>This value is PER CHAIN Power for each enabled chain</u>** |
| | Use this field to enter the desired power for current test case. The test lab will generally confirm that the average conducted power is within a small margin of this value using a power meter, before commencing testing. Samples are expected to transmit with conducted power within 1 dB of the value input above. | |



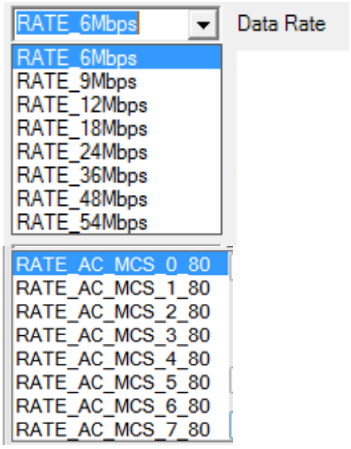
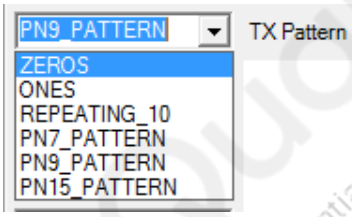
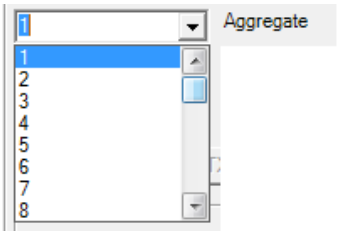
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| Step | QRCT Menu Items | Options |
|---|---|---|
| **5** | HT mode<br>  | **Capability Bandwidth (CBW) determined by this setting.**<br>Testing should be performed using the widest supported CBW supported by the product – regardless of the tested Packet Bandwidth (PBW) set in Chapter 6 – Data Rate)<br><br>20 MHz BW:<br>▪ No HT: No channel bonding. For 802.11a/g testing.<br>▪ HT20: 20 MHz bandwidth 802.11n<br>▪ CCK: CCK bandwidth. For 802.11b testing.<br><br>40 MHz BW:<br>▪ HT40+: 40 MHz bandwidth, primary high, 802.11n<br>▪ HT40: 40 MHz bandwidth, primary low, 802.11n<br>▪ VHT20: 20 MHz bandwidth 802.11ac<br>▪ VHT40+: 40 MHz bandwidth, primary high, 802.11ac<br>▪ VHT40-: 40 MHz bandwidth, primary low, 802.11ac<br><br>80 MHz BW:<br>▪ VHT80_0: 80 MHz bandwidth 20/40 MHz low, 40/80 MHz low<br>▪ VHT80_1: 20/40 MHz low, 40/80 MHz high<br>▪ VHT80_2: 20/40 MHz high, 40/80 MHz low<br>▪ VHT80_3: 20/40 MHz high, 40/80 MHz high<br><br>80 MHz + 80 MHz BW (non-contiguous):<br>▪ BW80p80 :11ac non-contiguous 80+80 MHz mode<br>▪ Primary_(1st~8th)_20_in_80p80: 80+80 mode with different primary 20 MHz options.<br>▪ Primary_(1st~8th)_20_in_16080+80 mode with different primary 20 MHz options.<br>[When BW80p80 is selected, the 2nd channel frequency selection will be available in item #2 menu].<br><br>160 MHz BW (contiguous):<br>▪ BW160: 11ac 160 MHz mode. |
| | Choices shown in QRCT will be limited by the chip capability. Refer to the manufacturer-provided test plan for bandwidths and modes to be tested. | |

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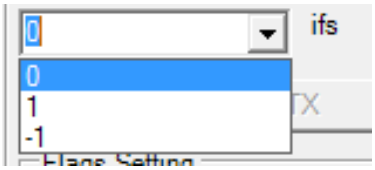
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| Step | QRCT Menu Items | Options |
|---|---|---|
| **6** | Data rate<br>  | All HT20/40/80/160 data rates depending on HT mode |
| | Choose Data Rate per selected mode in Step #5.<br>The lowest data rate (MCS0) (per each mode) transmits with the highest power. | |
| **7** | Tx pattern<br>  | ▪ ZEROS: Select default<br>▪ ONES : 111111<br>▪ Repeating_10 :101010101<br>▪ PN7_Pattern : 0111<br>▪ PN9_Pattern: 1001<br>▪ PN15_Pattern: 1111 |
| | Select “ZEROS”<br>Tx data Pattern represent digital domain data and does not impact conformance results.<br>There is no need to repeat conformance testing using different patterns. | |
| **8** | Short guard<br>  | ▪ Don’t use: Short guard Off<br>Use: On |
| | Short Guard and Long Guard has no effect on conformance result. | |
| **9** | Aggregate<br>  | 1 to 32: Number of PDU frame in a packet. |
| | This setting has no effect when using QRCT in TX99 Continuous Transmission mode. Default set to 1. | |

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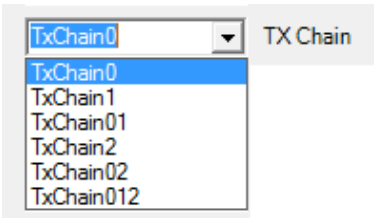
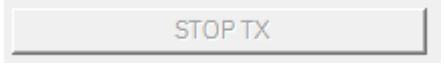
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| Step | QRCT Menu Items | Options |
|---|---|---|
| **10** | IFS (Interframe spacing)<br>  | ▪ 0<br>▪ 1<br>▪ -1 |
| | This setting has no effect when using QRCT in TX99 Continuous Transmission mode. Default set to 0. | |
| **11** | # of packets<br>  | 0: For continuous Tx |
| | Set to “0” for continuous TX mode. | |
| **12** | Scrambler<br>  | ▪ On: Scrambler turned on<br>▪ Off: Scrambler turned off |
| | This setting has no effect when using QRCT in TX99 Continuous Transmission mode. Default set to Off. | |
| **13** | ASIFN (arbitration interframe spacing timing parameter)<br>  | ▪ 0 to 252: Spacing timing between frames. |
| | This setting has no effect when using QRCT in TX99 Continuous Transmission mode. Default set to 1. | |
| **14** | Packet size<br>  | ▪ Size of packets (in bytes).<br>▪ This setting adjusts the length of the data portion of repeated packets when using Tx99 mode. However, the value has no impact on measured transmit power or emissions. |
| | Use default “1500” packets size | |
| **15** | Antenna<br>  | ▪ 0: Antenna 0<br>▪ 1: Antenna 1<br>Feature not supported for most current chips. No impact to compliance testing. This feature was previously used to switch one chain between two different antenna ports. |
| | Not currently used | |

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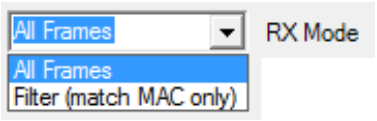
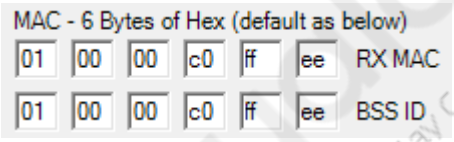
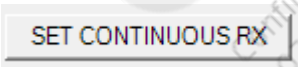
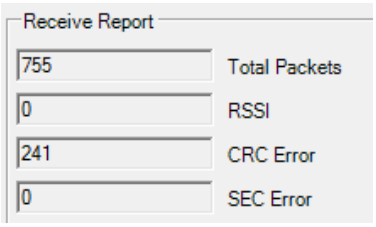
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| Step | QRCT Menu Items | Options |
|---|---|---|
| **16** | Tx chain<br>  | ▪ TxChain0<br>▪ TxChain1<br>▪ TxChan01<br>▪ TxChain2<br>▪ TxChain02<br>▪ TxChain012 |
| | Select which chain(s) will transmit (or receive) during testing:<br>Tx Chain0: TX using chain 0 only.<br>Tx Chain 01: TX using chain 1 only.<br>Tx Chain 02: TX using chains 0 and 2 only.<br>Tx Chain 0123: TX using chains 0,1,2 and 3 | |
| **17** | Gain index<br>  | ▪ 0 to XX (0-31) |
| | Only applicable when setting forcedgain in #3. Not used for conformance testing. | |
| **18** | DAC gain<br>  | ▪ 0 to XX (0-64) |
| | Only applicable when setting forcedgain in #3. Not used for conformance testing. | |
| **19** | PA CFG (PA configuration)<br>  | |
| | Only applicable when setting forcedgain in #3. Not used for conformance testing. | |
| **20** | Flags setting<br>  | ▪ Enable LDPC<br>▪ Enable STBC<br>▪ Enable DPD<br>▪ Enable HeaveClip |
| | LDPC: No effect on conformance testing. Leave unchecked.<br>STBC: Not used for conformance testing. Leave unchecked.<br>DPDmod: Not used for conformance testing. Leave unchecked. | |
| **21** | Click **SET TX ON** to begin transmission.<br>  | Click to start TX. |
| | Note that all parameters should be confirmed and set before “SET TX ON” | |
| **22** | Click **STOP TX** to stop transmission.<br>  | Click to stop TX |
| | Press “STOP TX” before making changes to the above settings. | |



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### Table 10-2 QRCT WLAN receiver settings and testing [for ETSI]

| Step | QRCT Menu Items | Options |
|---|---|---|
| 1 | Rx mode<br>  | ▪ All Frames: Rx all frames<br>▪ Filter (match MAC Only): Rx only frames matching MAC address |
| | Select “All Frames” | |
| 2 | Anti-noise<br>  | ▪ On<br>▪ Off |
| | Does not affect regulatory conformance results | |
| 3 | MAC – 6 bytes of hex<br>  | ▪ 0: Default antenna<br>▪ 1: Antenna 1 |
| | No impact to regulatory conformance results.<br>Applicable only when “Filter (match MAC Only)” in #1 is selected. | |
| 4 | Click **SET CONTINUOUS RX**.<br>  | |
| | Click to start continuous RX. | |
| 5 | Use VSG to send packets. | ▪ Set Rx MAC<br>▪ Set BSS ID MAC |
| | Does not affect regulatory conformance results. | |
| 6 |  | ▪ Click GET RECEIVE REPORT. |
| | Does not affect regulatory conformance results. | |
| 7 |  | Inspect Receive Report and calculate PER. |
| | Does not affect regulatory conformance results. | |



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